

7	(a) for a wave, electron can ‘collect’ energy continuously for a wave, electron will always be emitted / electron will be emitted at all frequencies..... after a sufficiently long delay	B1 M1 A1	[3]
(b)	(i) <i>either</i> wavelength is longer than threshold wavelength <i>or</i> frequency is below the threshold frequency <i>or</i> photon energy is less than work function	B1	[1]
	(ii) $hc / \lambda = \phi + E_{\text{MAX}}$ $(6.63 \times 10^{-34} \times 3.0 \times 10^8) / (240 \times 10^{-9}) = \phi + 4.44 \times 10^{-19}$ $\phi = 3.8 \times 10^{-19} \text{ J (allow } 3.9 \times 10^{-19} \text{ J)}$	C1 C1 A1	[3]
(c)	(i) photon energy larger so (maximum) kinetic energy is larger	M1 A1	[2]
	(ii) fewer photons (per unit time) so (maximum) current is smaller	M1 A1	[2]
8	(a) (i) E_0 shown near peak	A1	[1]